

Semantic Gap Analysis: An NLP & Mathematical Intent Classifier for Cosmo

Technical Report

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Date: July 2025

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1. Introduction

1.1 Background: Cosmo's Architecture and Philosophy

Cosmo is a conversational interface built to explore the balance between transparency and capability in AI. Rather than relying on large language models, Cosmo uses a hand-curated knowledge base stored in Firebase Firestore. Every response was manually written, edited, and structured with over 100 Q&A pairs.

The philosophy behind Cosmo emerged from a specific tension observed while navigating the tech industry's "no degree, no experience" paradox. There is an implicit pressure to present as omniscient—to mask uncertainty, hide gaps, and perform expertise. Cosmo is a deliberate counterpoint to that pressure, embodying the principle that purposeful work does not require perfection.

Cosmo's architecture reflects these philosophical commitments. He is honest by design, explicitly stating when he lacks information. He acknowledges his limitations rather than simulating omniscience. He is manually curated and human-paced to ensure accuracy and alignment with my actual experience.

However, this handcrafted approach created a practical problem: users frequently ask the same question in different ways, and Cosmo sometimes fails to recognize variations not explicitly encoded in the knowledge base. The challenge is to expand his capability while staying true to his design principles.

1.2 Problem Statement

Cosmo's retrieval engine relies on exact matching and keyword scoring against a set of stored variations for each document. When a user's question does not match any stored variation, Cosmo either returns no response or retrieves an incorrect answer.

The core challenge: how can I systematically identify which semantic variations are missing from the knowledge base without manually testing every possible question formulation? This report presents a diagnostic methodology to measure semantic coverage, identify blind spots, and guide targeted knowledge base expansion.

1.3 Research Questions

This analysis addresses three questions. First, how well does Cosmo's current knowledge base capture semantic variations of user intents? Second, can vector-based semantic similarity methods such as TF-IDF and LSA identify systematic blind spots? Third, what magnitude of improvement does LSA offer over baseline term frequency matching?

2. Methodology

2.1 Data Extraction and Preprocessing

The dataset consists of 139 documents exported from Firebase Firestore. Each document contains a primary question, an answer, an array of alternative variations, suggested followup questions, a priority weight, and keywords. From these documents, 680 total variations were extracted, yielding a vocabulary of 602 unique words.

A preprocessing pipeline was implemented to prepare the text for vectorization. The `clean_text()` function converts text to lowercase, removes possessives, strips punctuation while retaining alphanumeric characters, splits into tokens, removes stopwords from a custom list of over 80 common words, and filters out single-character tokens. This preprocessing ensured that vectorization captured meaningful semantic content while discarding noise.

2.2 Query Generation

To systematically test semantic coverage, I generated synthetic queries from each document's topics. Topics were extracted from the primary question, the alternative variations (up to 5 for high-priority documents and 3 for others), and the followup questions (up to 2 for high-priority documents and 1 for others).

The generation process used three methods. A subset of five question templates from a library of fifteen were applied to each topic, including formulations like "what is {topic}", "tell me about {topic}", "how does {topic} work", and "what do you know about {topic}". A manually curated synonym map with over 40 mappings enabled substitution of key terms with semantic equivalents—for example, "management" could become "leadership", "supervision", or "direction". Additional conversational variations were generated, including "I'd like to know about {topic}" and "can you tell me more about {topic}".

This process generated 7,800 total unique queries. All 7,800 queries were analyzed to ensure complete coverage, no sampling was used in the final methodology.

2.3 Vectorization Strategies

I implemented and compared three vectorization methods.

Term Frequency (TF) was implemented as the baseline method. TF represents each document as a vector where each token's value equals its normalized frequency within the document: $TF(t, d) = \text{count}(t, d) / \text{total_tokens}(d)$. This approach preserves frequency information and is computationally efficient. Moreover TF captures how many times a term appears, providing richer semantic information.

TF-IDF weights terms by their discriminative power across the corpus. The formula $TF\text{-}IDF(t, d, D) = TF(t, d) \times \log(|D| / (1 + df(t, D)))$ reduces the weight of common terms and increases the weight of distinctive terms. I computed IDF across all 680 variations and applied it to the TF vectors.

Latent Semantic Analysis (LSA) applies Singular Value Decomposition (SVD) to the TF-IDF matrix to discover latent semantic structure. The decomposition $X = U \times \Sigma \times V^T$ produces document-concept relationships, concept weights, and concept-term relationships. Dimensionality was reduced by selecting the top k singular values. Rather than using a fixed variance threshold, I tested multiple k values and selected the one that maximized performance. All k values were tested on all 7,800 queries (not just a sample) to ensure robust selection. This approach identified $k = 13$ as optimal, capturing only 9.6% of variance but achieving the highest similarity scores. This finding aligns with established LSA literature, which often finds that the first few dimensions capture the most important semantic structure while additional dimensions introduce noise.

2.4 Similarity Measurement

I used cosine similarity to measure semantic alignment between experimental queries and baseline variations. The formula $\text{cosine}(a, b) = (a \cdot b) / (||a|| \times ||b||)$ computes the cosine of the angle between two vectors, with higher scores indicating stronger semantic alignment.

For each experimental query, I found the baseline variation with maximum cosine similarity. Scores were interpreted using heuristic thresholds: scores above 0.7 indicate strong alignment where the query would likely match correctly, scores between 0.4 and 0.7 indicate moderate alignment where the query might match but less reliably, and scores below 0.4 indicate weak alignment where the query likely fails in retrieval.

2.5 Evaluation Framework

To validate that similarity scores reflect genuine semantic signal rather than random chance, I generated random vectors and computed their maximum cosine similarity with any baseline variation. This established the "noise floor" for comparison.

Each query was traced back to its source document using a multi-stage matching process: direct substring matching, Jaccard similarity with a 0.3 threshold, and word-level matching. This ensured thorough coverage analysis. Similarity scores were analyzed for distribution patterns including mean, median, standard deviation, and alignment category counts. Results were exported as both JSON and CSV for further analysis and sharing.

Statistical validation included 95% confidence intervals for all mean similarity scores and paired t-tests comparing TF and LSA performance.

2.6 Design Decisions

Choice of Optimal k for LSA. Rather than preselecting a variance threshold, I tested multiple k values and selected the one that maximized performance. This approach identified $k = 13$ as optimal, capturing only 9.6% of variance. This finding is consistent with LSA literature, which often finds that the first few dimensions capture the most important semantic structure while additional dimensions introduce noise. The 80% variance threshold commonly used in information retrieval would have selected $k = 249$, which actually performed worse (0.8570 vs 0.9665 for $k=13$).

Absence of Train-Test Split. This analysis does not employ a train-test split because it is diagnostic rather than predictive. The goal is to measure the semantic alignment between existing knowledge base entries and potential user queries, not to build a model that generalizes to unseen data. Since the queries are generated from the same documents they are tested against, a train-test split would not be meaningful—it would simply measure within-sample performance. The random baseline serves as the appropriate validation mechanism, confirming that observed alignment scores are significantly higher than chance. For predictive applications (e.g., building a classifier for future queries), a train-test split would be essential.

3. Results

3.1 Random Baseline Validation

The random baseline achieved an average score of 0.1074. In contrast, the TF vectors achieved an average score of 0.7967, representing a 641.9% improvement over random chance. This confirms that TF vectors contain genuine semantic information and that the observed scores are not artifacts of random similarity.

3.2 Baseline Performance (TF)

The Term Frequency approach produced an average cosine similarity of 0.7967 across all 7,800 experimental queries, with a median of 0.8165 and a standard deviation of 0.1558. The minimum score was 0.0000 and the maximum was 1.0000.

In terms of alignment categories, 6,152 queries or 78.9% showed strong alignment with scores above 0.7. One thousand five hundred fifty-nine queries or 20.0% showed moderate alignment with scores between 0.4 and 0.7. Only 89 queries or 1.1% showed weak alignment with scores below 0.4.

The 95% confidence interval for the TF mean was [0.7925, 0.8009], indicating high precision in the estimate. The margin of error for the strong alignment percentage was ± 0.9 percentage points, with a 95% confidence interval of [78.0%, 79.8%].

3.3 LSA Performance

With $k = 13$ dimensions (capturing 9.6% of variance), the LSA approach produced an average cosine similarity of 0.9665, representing a 21.3% relative improvement over the TF baseline. The median was 0.9998 and the standard deviation was 0.1210.

The alignment distribution shifted significantly. Strong alignment increased to 7,688 queries or 98.6%, an absolute improvement of 19.7 percentage points. Moderate alignment decreased to 8 queries or 0.1%, a reduction of 19.9 percentage points. Weak alignment increased slightly to 104 queries or 1.3%, an increase of 0.2 percentage points.

The 95% confidence interval for the LSA mean was [0.9639, 0.9692]. A paired t-test comparing LSA and TF scores yielded a t-statistic of -84.19 with $p < 0.001$, confirming that the improvement is statistically significant.

3.4 Document-Level Coverage

The analysis covered 136 out of 139 documents, or 97.8% of the knowledge base. The three documents that did not yield topics were default or generic entries such as "default," "hello," and "who are you," which were intentionally skipped during topic extraction to focus on substantive content. A total of 794 unique topics were extracted, generating 7,800 queries. The analysis of all 7,800 queries ensured complete coverage across all substantive documents.

3.5 Weak Alignment Analysis

The 89 queries with weak alignment under the TF approach were examined to understand the nature of the gaps. Representative examples include:

- **"how many people has niha managed"** — matched best with "management skills" at a score of 0.3391, indicating a missing quantitative intent in the variations.
- **"budget responsibility?"** — matched best with "leadership experience" at 0.2880, suggesting that this followup question lacked sufficient context in the knowledge base.
- **"what's Niha's philosophy on ethics"** — matched best with "ethical approach" at 0.3274, revealing that philosophical framing was absent from the existing variations.

Each weak alignment represents an actionable addition to the knowledge base.

3.6 Performance by Query Type

To understand where LSA provides the most benefit, I analyzed performance by query type. Synonym-heavy queries—those containing words from the synonym map, improved from 0.7525 (TF) to 0.9667 (LSA), a 28.5% improvement. Direct queries improved from 0.8264 to 0.9664, a 16.9% improvement.

This confirms that LSA is especially effective at capturing semantic variations introduced by synonyms.

3.7 Performance by Topic Category

Analysis by topic category revealed variation in LSA's effectiveness:

Category	Count	TF Score	LSA Score	Improvement
Management	421	0.7417	0.9708	+30.9%
Career	1,669	0.7580	0.9722	+28.3%
Projects	1,811	0.7540	0.9630	+27.7%
Skills	413	0.7583	0.9668	+27.5%
Education	707	0.7821	0.9658	+23.5%

Analysis by topic category revealed variation in LSA's effectiveness. Management queries benefited most (+30.9%) from LSA's ability to capture synonym relationships. Education queries benefited least (+23.5%), suggesting they were already well-covered by existing variations or had less semantic variation.

4. Discussion

4.1 Key Findings

Semantic coverage is exceptionally strong with LSA. With 98.6% of queries achieving strong alignment using LSA (versus 78.9% with TF), the knowledge base demonstrates robust coverage of semantic variation when using vector-based retrieval. The random baseline comparison confirms this signal is genuine and not an artifact of the methodology.

LSA provides substantial improvement. The 21.3% relative improvement from LSA is statistically significant ($p < 0.001$). LSA's advantage stems from capturing latent semantic relationships—words that do not co-occur but share conceptual similarity. For example, "supervisory experience" and "management background" are closer in LSA space than in TF space despite having limited lexical overlap.

The optimal $k = 13$ dimensions challenge the common 80% variance heuristic. While 80% variance ($k = 249$) achieved 0.8570, $k = 13$ (9.6% variance) achieved 0.9665. This demonstrates that the first few dimensions capture the most important semantic

structure, while additional dimensions introduce noise rather than signal. This finding aligns with established LSA literature.

The weak alignment rate under TF (1.1%) is manageable, and each weak alignment can be manually investigated and addressed with targeted variations. Document coverage was nearly complete, with 136 of 139 documents represented in the analysis.

4.2 Implications for Cosmo

The findings suggest three paths forward.

Immediate path (recommended): targeted expansion of the knowledge base using the weak alignment results to identify specific gaps and add 2-3 new variations per problematic document. This approach requires no code changes and leverages the existing retrieval system. It is the most realistic and immediately actionable path.

Hybrid path: using the LSA analysis to identify gaps, patching them with targeted variations, and monitoring performance over time. This approach maintains the current system while incrementally improving coverage.

Architectural path: integrating LSA to replace exact-match and keyword retrieval with LSA-based similarity, which would capture more semantic variation automatically while preserving transparency and factual accuracy.

The optimal $k = 13$ finding suggests that a lightweight LSA implementation with only 13 dimensions would be computationally efficient while delivering near-maximum performance. This is particularly relevant for Cosmo's client-side JavaScript implementation, where computational resources are limited.

4.3 Limitations

The synthetic queries generated for this analysis may not perfectly represent real user language, though the multi-method generation likely covers the most common alternative phrasings. The analysis is specific to Cosmo's knowledge base, and generalizability to other domains or larger datasets would require further study.

The heuristic thresholds used for alignment categories (0.7 for strong, 0.4 for moderate) are standard in information retrieval literature and validated by the random baseline, which confirms that observed scores are significantly higher than chance. However, these thresholds would benefit from user testing to establish optimal values for actual retrieval performance.

The analysis uses cosine similarity to predict retrieval accuracy but does not measure actual user satisfaction or task completion. Since Cosmo serves as a conversational portfolio where users engage with personal narrative and professional experience, satisfaction depends on factors beyond retrieval accuracy—including

tone, authenticity, and perceived helpfulness.

User testing would be required to validate that improved retrieval correlates with improved user experience.

Finally, while the analysis achieved 97.8% document coverage, the three documents that were skipped represent edge cases where topic extraction was intentionally limited to substantive content.

5. Conclusion and Future Work

This analysis presents a systematic methodology for identifying semantic blind spots in retrieval-based conversational systems. The key findings demonstrate that baseline TF vectors achieve strong alignment with an average score of 0.7967, representing a 641.9% improvement over random chance. LSA provides a 21.3% relative improvement with an average score of 0.9665, using only 13 latent dimensions that capture 9.6% of the variance in the TF-IDF matrix. Only 1.1% of queries show weak alignment under TF, representing actionable gaps for knowledge base expansion.

The diagnostic nature of this analysis justifies the absence of a train-test split—the goal is measurement and identification of gaps, not predictive modeling. The random baseline serves as the appropriate validation mechanism, confirming that observed alignment scores are significantly higher than chance.

For Cosmo, these findings validate the current approach while providing a clear path to improvement. The small number of weak alignments suggests that targeted, manual expansion may be sufficient to achieve near-complete semantic coverage without architectural changes. However, the dramatic improvement from LSA suggests that a vector-based retrieval system would significantly enhance Cosmo's ability to understand user queries.

Several directions for future work emerge from this analysis. Applying the same methodology to real user queries would prioritize variations based on actual usage patterns. A/B testing between exact-match and LSA retrieval with real users would provide empirical evidence of user experience improvement. Implementing automated weekly analysis would enable continuous monitoring of semantic coverage over time. Finally, extending the methodology to non-English variations would enable Cosmo to support multiple languages.

This analysis itself serves as a portfolio piece, demonstrating my ability to design and execute a rigorous technical investigation—from problem formulation through methodology design, implementation, statistical validation, and actionable recommendations. It reflects the same commitment to transparency and thoughtful engineering that Cosmo embodies.